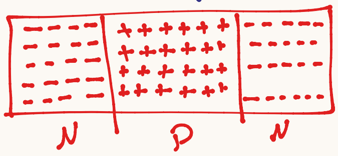
Electronic Switch

↳ it's electronics component → connect & disconnect between two point by electricity

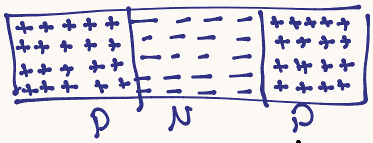
example ① transistor ② optocoupler ③ Relay

[1] transistor → BJT | MOSFET



* NPN

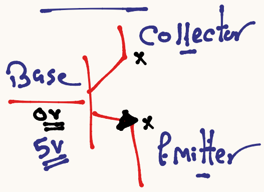
Use Case
① switching
② Amplification

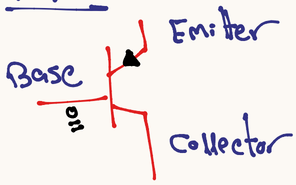


* PNP

* we will use transistor in switching phase?

NPN	PNP





Base = 0
↳ O.C

o.c | collector
| Emitter

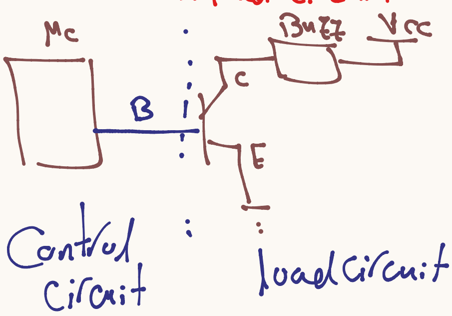
Base = 1
↳ S.C

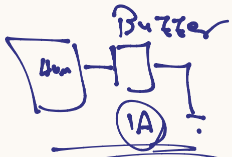
| C
| E

Base = 0 → PNP → SC

Base = 1 → PNP → O.C

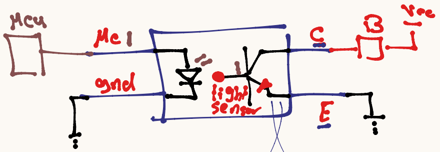
* transistor → Isolate between control circuit & load circuit





issue

* optocoupler



Fully Isolate between load Control.

* 4N35 → PNP
* PC817 → NPN

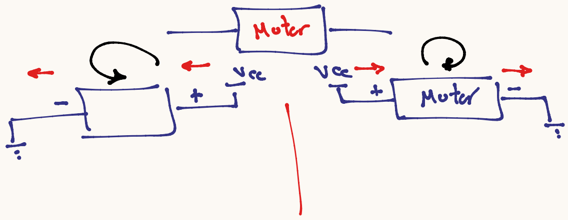
* How decrease for pin Control on Motor

* B_1 & B_3 → Mc → Q_1
* B_2 & B_4 → Mc → Q_2

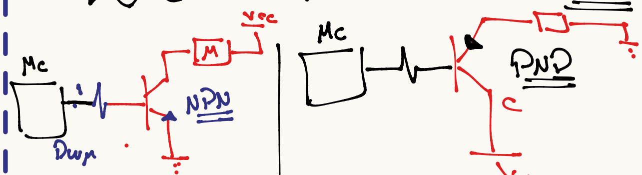
CW	CCW	off
$A_1=1$	$A_1=0$	$A_1=0$
$A_2=0$	$A_2=1$	$A_2=0$

* Motor
  ↳ DC Motor →
  ↳ stepper →
  ↳ servo →

DC Motor → Apply volt → work



* DC Connection
↳ ① Control on Motor by on/off only



1 → Motor on
0 → Motor off

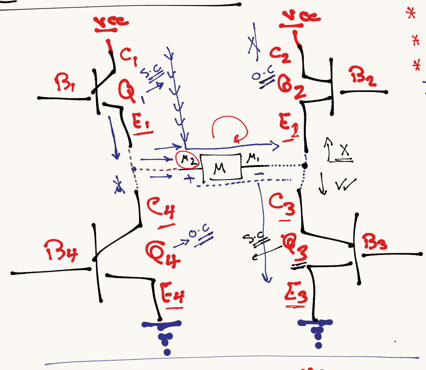
1 → Motor off
0 → Motor on.

* Note: this Connection Can be Control on Motor speed also but Must Connect with PWM Pin

* ② Control DC Motor by Direction
↳ ① off ↳ ② on CW
↳ ③ on CCW

* we need the H-bridge Circuit
① transistor
↳ 4 NPN ↳ 4 PNP ↳ 2 NPN 2 PNP

Case → 4 NPN

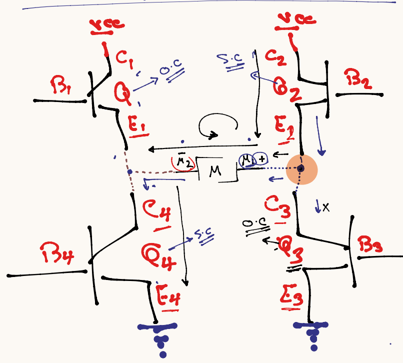


* C_1 & C_2 → 5V
* E_3 & E_4 → Gnd.
* E_1 & C_4 → M_2 (Motor).
* E_2 & C_3 → M_1 (Motor).

B → Mc
B_1 → A_1
B_2 → A_2
B_3 → A_3
B_4 → A_4

Case1 → CW

Q_1 → S.C $B_1=1$ $A_1=1$
Q_2 → O.C $B_2=0$ $A_2=0$
Q_3 → S.C $B_3=1$ $A_3=1$
Q_4 → O.C $B_4=0$ $A_4=0$



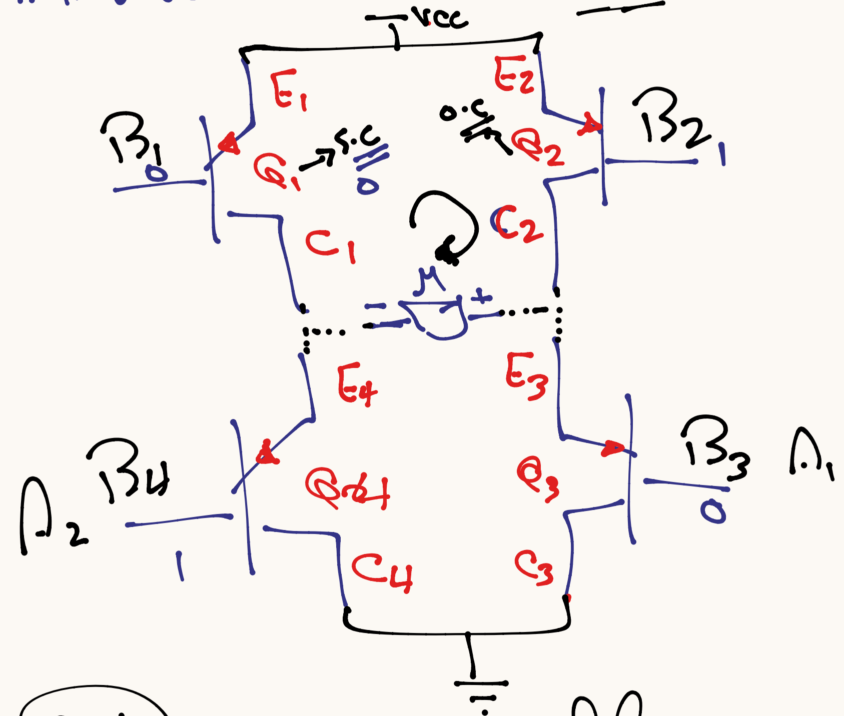
Case2 → CCW

Q_1 = O.C $B_1=0$ $A_1=0$
Q_2 = S.C $B_2=1$ $A_2=1$
Q_3 = O.C $B_3=0$ $A_3=0$
Q_4 = S.C $B_4=1$ $A_4=1$

Case 3 → off

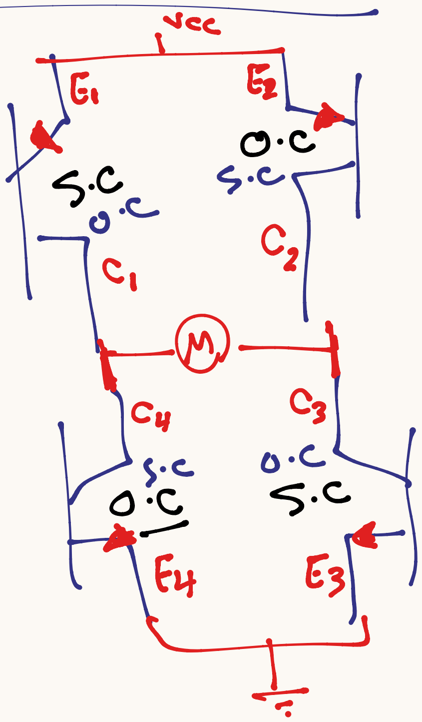
Q_1 O.C $B_1=0$ $A_1=0$
Q_2 O.C $B_2=0$ $A_2=0$
Q_3 O.C $B_3=0$ $A_3=0$
Q_4 O.C $B_4=0$ $A_4=0$

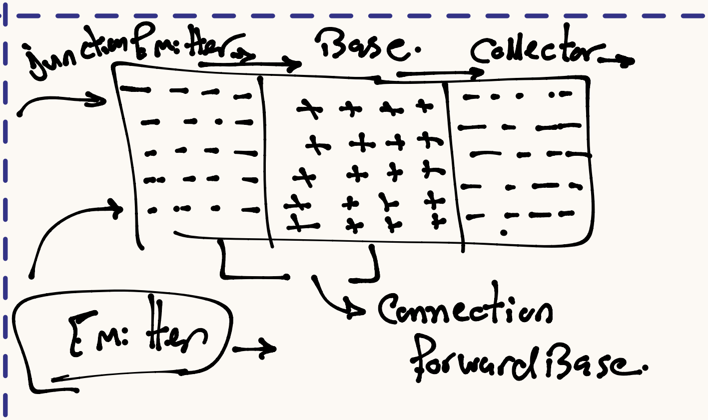
* H-bridge → 4 transistor PNP



CW	CCW	off
$A_1 = 0$	$A_1 = 1$	$A_1 = 1$ (o.c)
$A_2 = 1$	$A_2 = 0$	$A_2 = 1$ (o.c)

2 NPN - 2 PNP





NPN → Emitter → 0v

↳ Collector → load
↳ Vcc

PNP → Emitter → Vcc

Collector → load
↳ Gnd

CW PNP

$B_1 \to 0$

$B_2 \to 1$

$B_3 \to 1$

$B_4 \to 0$

CCW

$B_1 \Rightarrow 1$

$B_2 \Rightarrow 0$

$B_3 \Rightarrow 0$

$B_4 \to 1$

NPN

		CW	CCW
$B_1 \to B_4$	→ A_1	0	1
$B_2\ B_3$	→ A_2	1	0